# The Standard Quantum Limit in Measurement Physics

A Scaffolded Treatise

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**Abstract**

The standard quantum limit is a family of precision benchmarks derived once a measurement model, resource allocation, probe state, readout mechanism, and noise convention are specified. In continuous position measurements, the conventional SQL emerges from equilibrating readout imprecision with quantum backaction for an uncorrelated meter. For phase estimation, it denotes the inverse square-root scaling inherent to independent probes. In phase-preserving amplification, it represents the minimal input-referred noise mandated by commutator preservation. Although these definitions share a unified information-theoretic framework, each addresses distinct experimental inquiries. This study advances the field from raw measurement records and spectral densities through linear response theory, quantum noise constraints, Fisher information analysis, to experimentally calibrated sub-SQL assertions. The derivations differentiate single-sided versus double-sided spectra, isolate imprecision from overall sensitivity, link continuous SQL measurements with constrained quantum Cramér–Rao bounds, and illustrate how squeezing, correlations, variational readouts, speed meters, backaction evasion techniques, entanglement, and quantum nondemolition observables modify the underlying assumptions that define the benchmark. The findings are applicable to a broad spectrum of technologies, including gravitational-wave interferometry, cavity optomechanics, quantum imaging, force microscopy, spin ensembles, atomic clocks, and microwave amplification.

# Contents

## 1 Scope, vocabulary, and historical calibration

**Before reading this section**

Ensure that expectation values, variances, canonical commutators, and elementary Fourier transforms are familiar. Later sections introduce density matrices and Fisher information before using them.

Precision measurement starts with a parameter $\theta$, a physical system that carries its influence, a transduction law, a classical record, and an estimator. Displacement, force, strain, phase, frequency, and magnetic field are common parameters. A mirror, resonator, optical mode, atomic ensemble, spin ensemble, or microwave mode can carry the response. The apparatus converts that response into voltages or photon counts, and an estimator maps the record to $\hat{\theta}$ with a calibrated uncertainty.

The phrase standard quantum limit acquires meaning after this chain has been specified. A conventional continuous position meter reaches its SQL at the balance between imprecision and backaction. An ensemble of independent probes reaches a statistical SQL with uncertainty proportional to $N^{-1/2}$. A phase preserving amplifier reaches an added noise limit fixed by bosonic commutators. These benchmarks can share a laboratory system, yet a numerical comparison requires matching the measured quantity, bandwidth, resource count, and spectral convention [1–3].

**Table 1:** Four uses of quantum limit language that require separate assumptions.

| Setting | Bounded quantity | Assumptions that produce the standard benchmark | Common expression |
|---|---|---|---|
| Continuous position meter | Added displacement or force spectrum | Linear response, minimum uncertainty meter, and zero useful imprecision and backaction correlation | $\bar{S}_{xx}^{\mathrm{SQL}} = \hbar\lvert\chi_m\rvert$ |
| Statistical phase estimation | Estimator standard deviation | Independent probes, additive generator, and fixed single probe resource | $\Delta\theta \propto N^{-1/2}$ |
| Phase preserving amplifier | Input referred added noise | Deterministic linear gain for both field quadratures | $n_{\mathrm{add}} \geq 1/2$ at large gain |
| Correlation aware force sensing | Added force spectrum | Stationary linear sensing with optimized correlations and dissipative response | $\bar{S}_{FF}^{\mathrm{add}} \geq \hbar\lvert \operatorname{Im}\chi_m^{-1}\rvert$ |

The historical record developed in stages. Braginskii analyzed classical and quantum restrictions on detecting weak disturbances of a macroscopic oscillator in work published in Russian in 1967 and in English translation in 1968 [1]. Braginsky and Vorontsov then formulated macroscopic quantum measurement limits, and Braginsky and Manukin assembled weak force detection methods into a monograph [4, 5]. Work on repeatable oscillator observables and weak force estimation led to quantum nondemolition measurement, whose gravitational wave motivation was presented by Braginsky, Vorontsov, and Thorne and developed in a detailed review by Caves and collaborators [2, 6, 7].

Caves separated photon counting uncertainty from radiation pressure fluctuations in interferometers, introduced squeezed input as a route beyond coherent state performance, and established the phase preserving amplifier noise theorem [8–10]. Later reviews placed these results inside linear response, cavity optomechanics, and modern gravitational wave detector theory [3, 11–15].

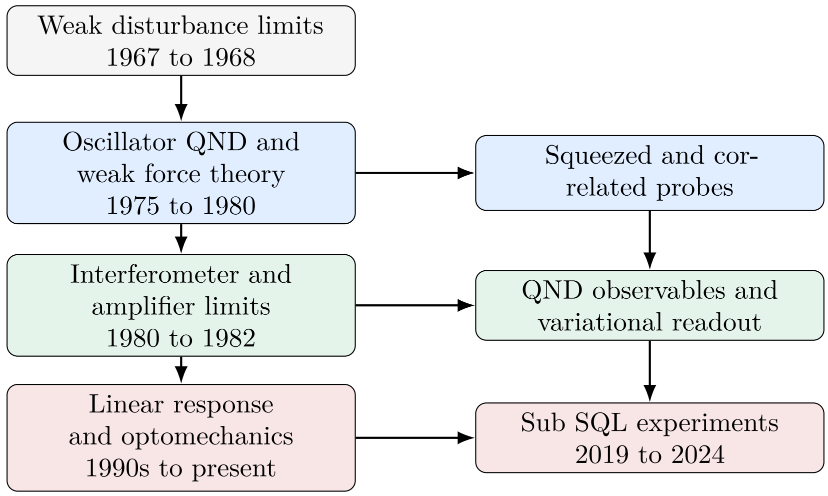


**Figure 1:** Conceptual lineage of the SQL. The arrows indicate dependence among ideas and carry no claim of exclusive authorship.

**Table 2:** Core vocabulary used throughout the article.

| Term | Meaning |
|---|---|
| Imprecision | Noise referred to the input that obscures the measured variable in the readout record. Optical phase fluctuations and amplifier output noise are common sources. |
| Backaction | Fluctuating disturbance exerted by the meter on the measured system. Radiation pressure force and charge force noise are common examples. |
| Susceptibility | Complex linear response $\chi_m(\Omega)$ that maps applied force to displacement. Its magnitude sets signal gain, and its phase sets the correlation needed for cancellation. |
| Spectral density | Frequency resolved covariance of stationary fluctuations. Its numerical value depends on sidedness, symmetrization, and Fourier normalization. |
| Coherent state | Minimum uncertainty optical state whose two quadratures have equal vacuum variance. It supplies the common laser reference. |
| Squeezed state | State with reduced variance in one field quadrature and increased variance in its conjugate quadrature. |
| Quantum nondemolition measurement | Repeated measurement of an observable whose future values remain compatible with earlier measurements under the relevant dynamics and coupling. |
| Backaction evasion | Coupling or readout design that directs disturbance away from the signal observable or cancels its contribution in the estimator. |
| Quantum Fisher information | Largest local Fisher information available from a parameter dependent quantum state after optimization over quantum measurements. |

**Practical rule of thumb**

Write the measured variable, resource, probe state, readout, bandwidth, and spectral convention beside every SQL formula. An unlabeled SQL curve carries incomplete physical information.

# 2 Measurement records, response functions, and spectra

**Before reading this section**

Ensure that complex Fourier amplitudes and linear harmonic response are familiar. The symmetrized spectral density is defined below, so prior experience with quantum noise spectra is unnecessary.

## 2.1 From a physical disturbance to an estimator

A linear sensor can be organized as a sequence of transduction stages. An external force changes a mechanical coordinate, an optical or electrical probe acquires that change, a detector produces a classical record, and a filter converts the record into an estimate. Each stage contributes gain, bandwidth, loss, and noise. Referring every contribution to the same input variable allows their covariances to be combined.

Let $x(t)$ denote the displacement of a mechanical mode and $F_{\mathrm{sig}}(t)$ the force to be estimated. The Fourier domain response is

$$x(\Omega) = \chi_m(\Omega)\left[F_{\mathrm{sig}}(\Omega) + F_{\mathrm{ba}}(\Omega) + F_{\mathrm{th}}(\Omega)\right], \tag{1}$$

where $F_{\mathrm{ba}}$ is meter backaction and $F_{\mathrm{th}}$ represents environmental force noise. For an oscillator with effective mass $m$, resonance $\Omega_m$, and energy damping rate $\Gamma_m$,

$$\chi_m(\Omega) = \frac{1}{m\left(\Omega_m^2 - \Omega^2 - i\Gamma_m\Omega\right)}. \tag{2}$$

The free mass approximation follows from $\Omega_m = 0$ and negligible damping over the signal band, which gives $\chi_m = -1/(m\Omega^2)$.

After calibration to displacement, the measured record is

$$y(\Omega) = x(\Omega) + x_{\mathrm{imp}}(\Omega), \tag{3}$$

where $x_{\mathrm{imp}}$ is input referred imprecision. Equations (1) and (3) form the minimal model behind the continuous measurement SQL.

## 2.2 Symmetrized double sided spectra

For stationary Hermitian fluctuations $\delta\hat{A}$ and $\delta\hat{B}$, this article uses

$$\bar{S}_{AB}(\Omega) = \frac{1}{2}\int_{-\infty}^{\infty} \mathrm{d}t\, e^{i\Omega t}\left\langle \left\{\delta\hat{A}(t), \delta\hat{B}(0)\right\}\right\rangle. \tag{4}$$

The bar marks a symmetrized double sided spectrum per radian per second. For an even classical spectrum, a one sided spectrum defined for $\Omega > 0$ satisfies

$$S_{AB}^{(1)}(\Omega) = 2\bar{S}_{AB}(\Omega). \tag{5}$$

An amplitude spectral density is the square root of a power spectral density. Displacement amplitude spectra carry units of $\mathrm{m}/\sqrt{\mathrm{Hz}}$, force amplitude spectra carry $\mathrm{N}/\sqrt{\mathrm{Hz}}$, and strain amplitude spectra carry $1/\sqrt{\mathrm{Hz}}$ after conversion to spectra per hertz.

**Convention checkpoint**

Every factor of two in an SQL spectrum depends on sidedness and symmetrization. Equations in this article use $\bar{S}$ for symmetrized double sided spectra. A one sided result follows from Eq. (5).

The meter covariance at each frequency can be represented by

$$\mathbf{S}_m(\Omega) = \begin{pmatrix} \bar{S}_{xx}^{\mathrm{imp}} & \bar{S}_{xF} \\ \bar{S}_{xF}^{*} & \bar{S}_{FF}^{\mathrm{ba}} \end{pmatrix}. \tag{6}$$

Input referral of the measurement record gives the added displacement spectrum

$$\bar{S}_{xx}^{\mathrm{add}}(\Omega) = \bar{S}_{xx}^{\mathrm{imp}} + |\chi_m|^2 \bar{S}_{FF}^{\mathrm{ba}} + 2\,\mathrm{Re}\Big[\chi_m^{*}\bar{S}_{xF}\Big]. \tag{7}$$

Intrinsic oscillator motion and environmental noise produce the measured spectrum

$$\bar{S}_{xx}^{\mathrm{meas}} = \bar{S}_{xx}^{\mathrm{intrinsic}} + \bar{S}_{xx}^{\mathrm{add}} + \bar{S}_{xx}^{\mathrm{tech}}. \tag{8}$$

The cross spectrum $\bar{S}_{xF}$ is central. A phase readout of a coherent probe often has negligible useful correlation, whereas variational homodyne detection and ponderomotive squeezing can make the final term in Eq. (7) negative.

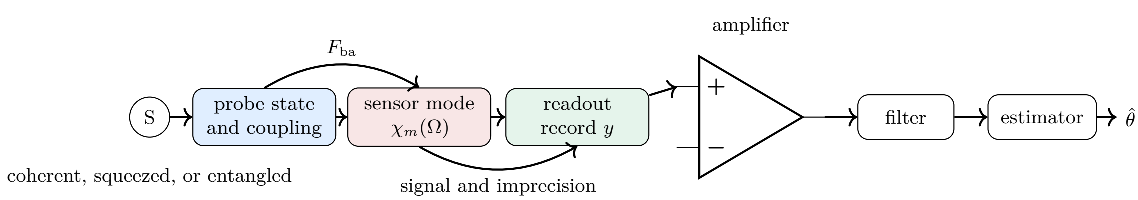


**Figure 2:** A calibrated quantum measurement chain. SQL performance depends on the full chain and on correlations retained by the estimator.

**Practical rule of thumb**

Refer every noise source to one variable before addition. Preserve the cross spectral terms until the chosen estimator and readout quadrature have been fixed.

## 3 Derivation of the continuous position and force SQL

**Before reading this section**

Review Eqs. (2), (6), and (7). The derivation uses a minimum uncertainty linear meter with negligible reverse gain and states every spectral convention explicitly.

### 3.1 Quantum noise constraint

A general linear detector has output operator $\hat{I}$, backaction operator $\hat{F}$, forward susceptibility $\chi_{IF}$, and reverse susceptibility $\chi_{FI}$. Define

$$\widetilde{\chi}_{IF}(\Omega) = \chi_{IF}(\Omega) - \chi_{FI}^{*}(\Omega). \tag{9}$$

The finite frequency quantum noise theorem can be written

$$\bar{S}_{II}\bar{S}_{FF} - |\bar{S}_{IF}|^2 \geq \left|\frac{\hbar\widetilde{\chi}_{IF}}{2}\right|^2 \left[1 + \Delta\left(\frac{\bar{S}_{IF}}{\hbar\widetilde{\chi}_{IF}/2}\right)\right], \tag{10}$$

where

$$\Delta[z] = \frac{|1 + z^2| - (1 + |z|^2)}{2}. \tag{11}$$

Because the correction $\Delta[z]$ vanishes in most standard quantum limit discussions, the subsequent sections transition directly to the simplified scalar inequality. This theorem retains gain, reverse gain, and finite frequency correlations [3].

For a canonical meter with negligible reverse gain, input referral by the forward gain and conditions that remove the correction reduce Eq. (10) to

$$\bar{S}_{xx}^{\mathrm{imp}}\bar{S}_{FF}^{\mathrm{ba}} - |\bar{S}_{xF}|^2 \geq \frac{\hbar^2}{4}. \tag{12}$$

The interaction Hamiltonian and output response are

$$\hat{H}_{\mathrm{int}} = -\hat{x}\hat{F}, \qquad \hat{I}(\Omega) = \hat{I}_0(\Omega) + \chi_{IF}(\Omega)\hat{x}(\Omega). \tag{13}$$

Equation (12) supplies the transparent algebra for the standard uncorrelated SQL. The full theorem is required for a general detector or a realizability proof [16, 17].

### 3.2 Optimization for an uncorrelated meter

Set $\bar{S}_{xF} = 0$ and parameterize measurement strength by a positive dimensionless variable $\lambda$,

$$\bar{S}_{xx}^{\mathrm{imp}} = \frac{A}{\lambda}, \qquad \bar{S}_{FF}^{\mathrm{ba}} = B\lambda. \tag{14}$$

Stronger coupling carries more information and lowers imprecision, and the same coupling raises the fluctuating force. The added displacement spectrum is

$$\bar{S}_{xx}^{\mathrm{add}} = \frac{A}{\lambda} + |\chi_m|^2 B\lambda. \tag{15}$$

Differentiation with respect to $\lambda$ gives

$$\lambda_{\mathrm{opt}} = \sqrt{\frac{A}{|\chi_m|^2 B}}, \tag{16}$$

and

$$\bar{S}_{xx,\mathrm{min}}^{\mathrm{add}} = 2|\chi_m|\sqrt{AB}. \tag{17}$$

For a meter that saturates Eq. (12), $AB = \hbar^2/4$, which yields

$$\boxed{\bar{S}_{xx}^{\mathrm{SQL}}(\Omega) = \hbar|\chi_m(\Omega)|}. \tag{18}$$

The force referred form follows from division by $|\chi_m|^2$,

$$\boxed{\bar{S}_{FF}^{\mathrm{SQL}}(\Omega) = \frac{\hbar}{|\chi_m(\Omega)|}}. \tag{19}$$

The imprecision and backaction contributions are equal at the optimum. This equality is a property of the optimum within the selected architecture and carries no universal requirement for every quantum sensor.

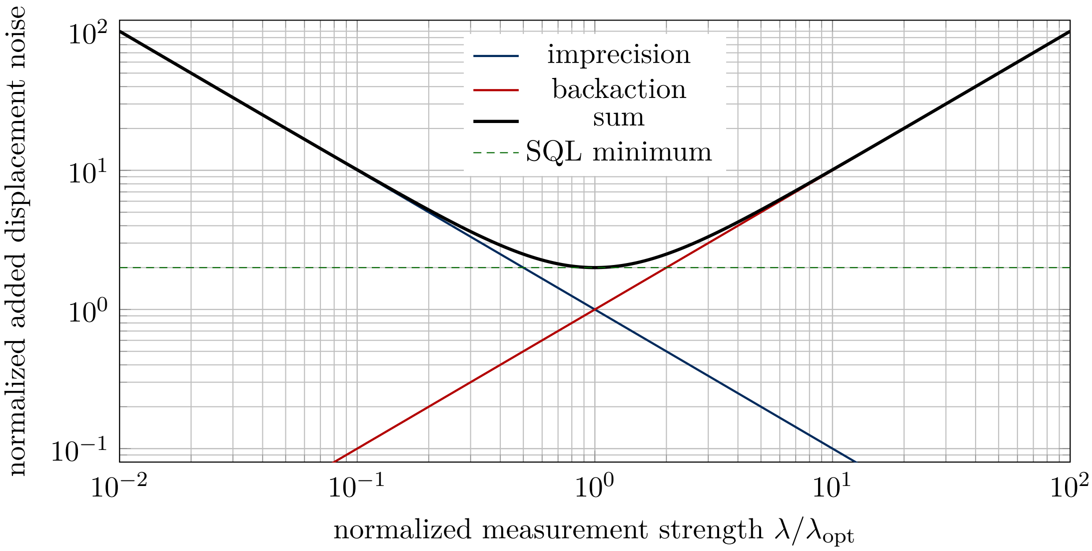


**Figure 3:** Normalized imprecision and backaction tradeoff for an uncorrelated linear meter.

## 3.3 Free mass and resonant oscillator

For a free mass, $|\chi_m| = 1/(m\Omega^2)$, so

$$\bar{S}_{xx}^{\mathrm{SQL}} = \frac{\hbar}{m\Omega^2}, \qquad \bar{S}_{FF}^{\mathrm{SQL}} = \hbar m\Omega^2. \tag{20}$$

For a lightly damped oscillator at resonance,

$$|\chi_m(\Omega_m)| = \frac{1}{m\Gamma_m\Omega_m}, \qquad \bar{S}_{FF}^{\mathrm{SQL}}(\Omega_m) = \hbar m\Gamma_m\Omega_m. \tag{21}$$

One sided spectra are twice these values. The square root amplitude spectra therefore acquire a factor $\sqrt{2}$.

The equilibrium one sided thermal force spectrum of a viscously damped oscillator can be written

$$S_{FF}^{\mathrm{th},(1)}(\Omega) = 2m\Gamma_m\hbar\Omega\coth\left(\frac{\hbar\Omega}{2k_BT}\right). \tag{22}$$

Its classical limit is $4m\Gamma_m k_BT$. Quantum measurement noise becomes experimentally visible after thermal, laser, electronic, and calibration contributions approach the relevant SQL spectrum.

## 3.4 Frequency dependent tuning landscape

The oscillator susceptibility changes rapidly near resonance, so a fixed probe strength reaches the optimum over a limited band. Define

$$\widetilde{\chi}(r) = \frac{1}{\sqrt{(1-r^2)^2 + (\gamma r)^2}}, \qquad r = \frac{\Omega}{\Omega_m}, \tag{23}$$

and normalized added noise

$$N(r,\lambda) = \lambda^{-1} + \lambda|\widetilde{\chi}(r)|^2. \tag{24}$$

With $u = \lambda|\widetilde{\chi}|$, the SQL ratio is

$$\frac{N}{N_{\mathrm{SQL}}} = \frac{1}{2}\left(u + u^{-1}\right) \geq 1. \tag{25}$$

Equality gives $\lambda_{\mathrm{opt}} = 1/|\widetilde{\chi}|$. Figure 4 evaluates this expression directly with PGFPlots and requires no external data file.

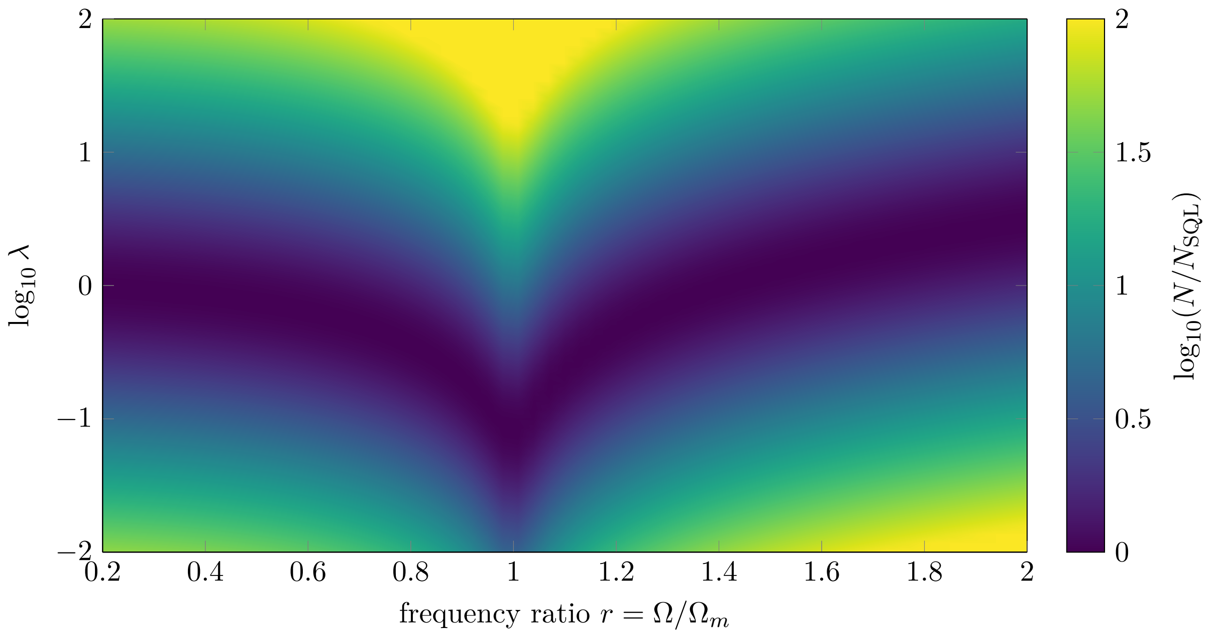


**Figure 4:** Excess added noise above the uncorrelated SQL for damping ratio $\gamma = 0.08$. The valley follows $\lambda_{\mathrm{opt}} = 1/|\widetilde{\chi}|$, and the plotted surface is generated analytically from the provided mathematical formulas to ensure computational reproducibility.

### 3.5 SQL and the correlation aware lower bound

Useful $\bar{S}_{xF}$ can lower Eq. (7) below Eq. (18). Stationary linear force sensing remains constrained by dissipation. In the same double sided convention, common forms of the ultimate added noise bounds are

$$\bar{S}_{xx}^{\mathrm{add}} \geq \hbar|\operatorname{Im}\chi_m|, \qquad \bar{S}_{FF}^{\mathrm{add}} \geq \hbar|\operatorname{Im}\chi_m^{-1}|. \tag{26}$$

For viscous damping, the force form is

$$\bar{S}_{FF}^{\mathrm{DQL}}(\Omega) = \hbar m \Gamma_m |\Omega|. \tag{27}$$

At mechanical resonance $\chi_m$ is nearly imaginary, so the uncorrelated displacement SQL and the dissipative displacement bound coincide. Away from resonance, their separation quantifies the improvement available from correlations. An ideal lossless free mass has a real susceptibility, and Eq. (26) then signals the importance of finite observation time, finite probe energy, optical loss, and other resource constraints that lie outside the stationary dissipative bound [18–20].

**Practical rule of thumb**

Balance imprecision and backaction to locate the uncorrelated SQL. Use the full covariance matrix and the dissipative response to evaluate any claimed advantage from correlations.

# 4 Optical interferometers and cavity optomechanics

**Before reading this section**

Ensure that optical phase, coherent state photon statistics, and harmonic oscillator susceptibility are familiar. Section 3 supplies the required spectral formulas.

## 4.1 Coherent light in a Michelson position meter

A differential arm displacement $x$ changes the round trip phase by

$$\phi = 2k_0 x = \frac{4\pi x}{\lambda_0}. \tag{28}$$

For $N$ detected photons in a coherent state, phase imprecision scales as

$$\Delta\phi_{\mathrm{imp}} \sim N^{-1/2}, \qquad \Delta x_{\mathrm{imp}} \sim \frac{\lambda_0}{4\pi\sqrt{N}}. \tag{29}$$

With optical power $P$, integration time $\tau$, and carrier frequency $\omega_0$, the detected photon number scales as $N \sim P\tau/(\hbar\omega_0)$. The displacement imprecision spectrum therefore scales as $P^{-1}$.

Coherent state amplitude quadrature fluctuations produce photon flux fluctuations around a bright carrier. Their fluctuating momentum transfer produces a radiation pressure force spectrum proportional to $P$. The phase quadrature produces readout imprecision. The equivalent displacement spectrum from force noise is multiplied by $|\chi_m|^2$, so a compact model is

$$\bar{S}_{xx}^{\mathrm{add}}(\Omega; P) = \frac{\alpha(\Omega)}{P} + \beta(\Omega) P |\chi_m(\Omega)|^2. \tag{30}$$

The optical commutator fixes $\alpha\beta$ for an ideal coherent probe. Minimization over $P$ reproduces Eq. (18). Strain calibration in a simple differential arm model uses $x = hL$, which gives $S_{hh} = S_{xx}/L^2$ after the effective differential mechanical mode has been defined.

The full interferometer input and output relations include arm cavities, signal recycling, photodetection loss, homodyne angle, and optomechanical coupling. Their frequency dependence rotates the output noise ellipse. Filter cavities can rotate an injected squeezed ellipse to track that response, which reduces high frequency phase noise and low frequency radiation pressure noise in one broadband configuration [21–23].

## 4.2 Cavity optomechanical implementation

For a single optical and mechanical mode, the dispersive Hamiltonian is

$$\hat{H} = \hbar\omega_c \hat{a}^\dagger \hat{a} + \hbar\Omega_m \hat{b}^\dagger \hat{b} - \hbar g_0 \hat{a}^\dagger \hat{a} \left(\hat{b} + \hat{b}^\dagger\right). \tag{31}$$

The single photon coupling $g_0$ gives the cavity frequency shift produced by zero point motion. Driving the cavity to mean photon number $n_c$ produces the linearized coupling $G = g_0\sqrt{n_c}$. For a resonant drive in the unresolved sideband measurement limit, a representative measurement rate is

$$\Gamma_{\mathrm{meas}} \simeq \frac{4\eta G^2}{\kappa}, \tag{32}$$

where $\eta$ is total detection efficiency and $\kappa$ is the cavity energy decay rate. Larger $n_c$ increases information rate and radiation pressure shot noise together. The cooperativity

$$C = \frac{4G^2}{\kappa\Gamma_m} \tag{33}$$

compares optomechanical measurement dynamics with mechanical damping. Quantum cooperativity also accounts for thermal occupation and indicates whether quantum backaction can compete with thermalization [14, 24].

The oscillator zero point displacement is

$$x_{\mathrm{zpf}} = \sqrt{\frac{\hbar}{2m\Omega_m}}. \tag{34}$$

For $k_B T \gg \hbar\Omega_m$, the thermal occupation is $n_{\mathrm{th}} \simeq k_B T/(\hbar\Omega_m)$. The product $\Gamma_m n_{\mathrm{th}}$ estimates a thermal heating rate. At fixed quality factor $Q = \Omega_m/\Gamma_m$,

$$\Gamma_m n_{\mathrm{th}} \simeq \frac{k_B T}{\hbar Q}. \tag{35}$$

Frequency alone therefore gives no reduction of this rate at fixed $Q$. Material loss, temperature, surface contamination, optical absorption, and mode shape determine the realized thermal environment.

Experiments have observed radiation pressure backaction, measured forces close to the SQL, and used optomechanical correlations to obtain total off resonant force and displacement sensitivity below the uncorrelated SQL [18, 25, 26]. The last statement is stronger than a report of imprecision below an SQL scale. It includes the calibrated thermal, intrinsic, imprecision, and backaction contributions in the measured band.

**Practical rule of thumb**

Increasing optical power improves phase readout until radiation pressure, absorption, heating, or dynamical backaction erases the gain. Track detected efficiency, intracavity power, and mechanical thermalization as separate control variables.

# 5 From continuous records to Fisher information

**Before reading this section**

Ensure that a density matrix $\rho$ is understood as a positive unit trace operator that describes pure and mixed quantum states. Review conditional probability and variance. This section defines every estimator and Fisher information quantity used later.

## 5.1 Classical estimators and local information

A measurement with outcomes $k$ and parameter dependent probabilities $p(k|\theta)$ has classical Fisher information

$$F_C(\theta) = \sum_k p(k|\theta) \left[\frac{\partial}{\partial\theta} \ln p(k|\theta)\right]^2. \tag{36}$$

For $\nu$ independent repetitions, a locally unbiased estimator obeys the Cramér–Rao inequality

$$\mathrm{Var}(\hat{\theta}) \geq \frac{1}{\nu F_C(\theta)}. \tag{37}$$

The bound describes local precision after the likelihood, measurement, and regularity conditions have been fixed. Bias, nuisance parameters, finite samples, broad priors, and periodic parameters require corresponding extensions.

A quantum state $\rho_\theta$ and a positive operator valued measure $\{E_k\}$ give

$$p(k|\theta) = \mathrm{Tr}(\rho_\theta E_k), \qquad E_k \succeq 0, \qquad \sum_k E_k = \mathbb{I}. \tag{38}$$

Optimization over all measurements yields quantum Fisher information. The symmetric logarithmic derivative $L_\theta$ satisfies

$$\partial_\theta \rho_\theta = \frac{1}{2}\left(L_\theta \rho_\theta + \rho_\theta L_\theta\right), \tag{39}$$

and

$$F_Q(\theta) = \mathrm{Tr}(\rho_\theta L_\theta^2), \qquad F_C(\theta) \leq F_Q(\theta). \tag{40}$$

The quantum Cramér–Rao bound is

$$\mathrm{Var}(\hat{\theta}) \geq \frac{1}{\nu F_Q(\theta)}. \tag{41}$$

Attainability depends on the state family, available measurement, local operating point, and multiparameter compatibility [27–29].

For unitary encoding generated by $\hat{G}$ and a mixed state $\rho = \sum_j p_j |j\rangle\langle j|$,

$$F_Q[\rho, \hat{G}] = 2 \sum_{j,k:p_j+p_k>0} \frac{(p_j - p_k)^2}{p_j + p_k} |\langle j|\hat{G}|k\rangle|^2. \tag{42}$$

For a pure state $|\psi\rangle$, this expression reduces to

$$F_Q = 4(\Delta\hat{G})^2_\psi. \tag{43}$$

Quantum Fisher information quantifies available information. Probe preparation, entanglement, squeezing, interaction engineering, and measurement design supply the physical resources that change it.

## 5.2 Independent probes and the statistical SQL

Let the parameter be encoded by

$$\hat{U}_\theta = e^{-i\theta\hat{G}}, \qquad \hat{G} = \sum_{j=1}^{N} \hat{g}_j. \tag{44}$$

For $N$ independent probes in a product state, local generator variances add,

$$(\Delta\hat{G})^2 = N(\Delta\hat{g})^2, \qquad F_Q \propto N, \qquad \Delta\theta \propto N^{-1/2}. \tag{45}$$

If the $j$th generator has spectral range $\Delta g_j = g_{j,\max} - g_{j,\min}$, every separable state obeys

$$F_Q[\rho_{\mathrm{sep}}, \hat{G}] \leq \sum_{j=1}^{N} (\Delta g_j)^2. \tag{46}$$

Identical qubits with $\hat{g}_j = \sigma_z^{(j)}/2$ satisfy $F_Q \le N$, so

$$\mathrm{Var}(\hat{\theta}) \ge \frac{1}{\nu N}. \tag{47}$$

This inequality gives a precise restricted state QCRB and supports the phrase statistical SQL [30, 31].

An ideal GHZ state places amplitude at the extremal eigenvalues of the total generator and reaches

$$F_Q = N^2, \qquad \mathrm{Var}(\hat{\theta}) \ge \frac{1}{\nu N^2}. \tag{48}$$

This Heisenberg scaling assumes a consistent resource count and coherent local encoding. Sequential coherent reuse can reproduce the same scaling under a resource definition that counts every pass. Independent loss and Markovian dephasing commonly return the asymptotic scaling toward $N^{-1/2}$, leaving a useful constant factor advantage over separable probes [32–34].

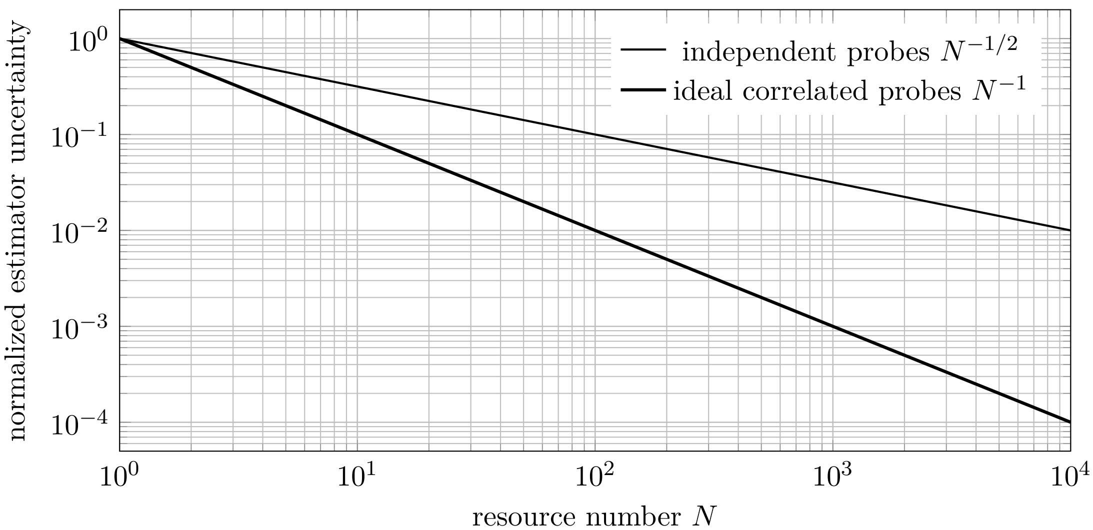


**Figure 5:** Ideal resource scaling. Loss, decoherence, and preparation overhead determine the realized advantage.

### 5.3 The bridge from spectra to estimators

The preceding continuous measurement theory describes a noisy record indexed by time or Fourier frequency. Fisher information describes the distinguishability of records or states indexed by a parameter. A stationary Gaussian record connects the two descriptions directly.

Consider

$$y(\Omega) = \theta s(\Omega) + n(\Omega), \tag{49}$$

where $s$ is a known finite duration signal template and $n$ is zero mean Gaussian noise with double sided spectrum $\bar{S}_{nn}$. The matched filter estimator is

$$\hat{\theta} = \frac{\displaystyle\int_{-\infty}^{\infty} \frac{\mathrm{d}\Omega}{2\pi} \frac{s^*(\Omega) y(\Omega)}{\bar{S}_{nn}(\Omega)}}{\displaystyle\int_{-\infty}^{\infty} \frac{\mathrm{d}\Omega}{2\pi} \frac{|s(\Omega)|^2}{\bar{S}_{nn}(\Omega)}}. \tag{50}$$

Its variance is

$$\mathrm{Var}(\hat{\theta}) = \left[ \int_{-\infty}^{\infty} \frac{\mathrm{d}\Omega}{2\pi} \frac{|s(\Omega)|^2}{\bar{S}_{nn}(\Omega)} \right]^{-1}, \tag{51}$$

which is the inverse Fisher information for this Gaussian model. Force estimation uses $s(\Omega) = \chi_m(\Omega) F_0(\Omega)$ after displacement calibration. Strain estimation uses the interferometer response to the chosen waveform.

Probe strength changes $\bar{S}_{nn}$ by changing imprecision, backaction, and their cross spectrum. Minimizing a narrowband noise spectrum therefore maximizes the Fisher information for a signal in that band. Broadband optimization weights every frequency by $|s|^2/\bar{S}_{nn}$, so the smallest pointwise spectrum and the smallest waveform estimation error can select different instrument settings. Functional quantum bounds extend this logic to time dependent forces and show how quantum noise cancellation and smoothing can approach the waveform QCRB [35]. The statistical and continuous SQLs meet at this estimator layer.

**Practical rule of thumb**

Use $F_Q \propto N$ as the precise separable probe benchmark. Use matched filter Fisher information for a continuous waveform. Keep resource counting, bandwidth, and estimator bias fixed during every comparison.

# 6 Physical routes below the standard benchmark

**Before reading this section**

Review the meter covariance matrix in Eq. (6), the added noise expression in Eq. (7), and the distinction between the SQL and Eq. (26).

## 6.1 Squeezed quadratures

Define optical quadratures by

$$\hat{X}_1 = \frac{\hat{a} + \hat{a}^\dagger}{2}, \qquad \hat{X}_2 = \frac{\hat{a} - \hat{a}^\dagger}{2i}, \qquad [\hat{X}_1, \hat{X}_2] = \frac{i}{2}. \tag{52}$$

Vacuum and coherent states have variance 1/4 in each quadrature. A squeezed Gaussian state has covariance

$$\mathbf{V}(r, \phi) = \frac{1}{4}\mathbf{R}(\phi) \begin{pmatrix} e^{-2r} & 0 \\ 0 & e^{2r} \end{pmatrix} \mathbf{R}^\top(\phi), \tag{53}$$

where $r$ is squeeze strength and $\phi$ is squeeze angle. A fixed phase squeezed input lowers high frequency readout imprecision and raises the conjugate amplitude fluctuations that drive low frequency radiation pressure. Frequency dependent rotation $\phi(\Omega)$ aligns the reduced variance with the frequency dependent signal quadrature [9, 21, 22].

Loss replaces a fraction of the state with vacuum,

$$\mathbf{V}_{\text{obs}} = \eta \mathbf{V}(r, \phi) + (1 - \eta)\frac{\mathbf{I}}{4}. \tag{54}$$

Large antisqueezing also converts small phase jitter into observed noise. Propagation loss, mode mismatch, photodetection efficiency, filter cavity loss, and squeeze angle control therefore set the realized improvement.

### 6.2 Correlation engineering and variational readout

Choose the phase of $\bar{S}_{xF}$ so that

$$\mathrm{Re}\Big[\chi_m^* \bar{S}_{xF}\Big] < 0. \tag{55}$$

The cross term in Eq. (7) then removes part of the backaction contribution from the record. Completing the square gives

$$\bar{S}_{xx}^{\mathrm{add}} = \left(\bar{S}_{xx}^{\mathrm{imp}} - \frac{|\bar{S}_{xF}|^2}{\bar{S}_{FF}^{\mathrm{ba}}}\right) + \bar{S}_{FF}^{\mathrm{ba}} \left|\chi_m + \frac{\bar{S}_{xF}}{\bar{S}_{FF}^{\mathrm{ba}}}\right|^2. \tag{56}$$

Within the reduced detector model, Eq. (12) gives

$$\bar{S}_{xx}^{\mathrm{add}} \geq \frac{\hbar^2}{4\bar{S}_{FF}^{\mathrm{ba}}} + \bar{S}_{FF}^{\mathrm{ba}} \left|\chi_m + \frac{\bar{S}_{xF}}{\bar{S}_{FF}^{\mathrm{ba}}}\right|^2. \tag{57}$$

Useful anticorrelation lowers the second term. The simplified determinant constraint alone leaves the broadband physical optimum undetermined. Detector gain, reverse gain, causality, dynamical backaction, optical loss, available coupling, and realizable homodyne angles constrain $\bar{S}_{xF}$.

Variational readout implements this idea by selecting a frequency dependent homodyne quadrature. Ponderomotive interaction creates correlations between optical amplitude and phase, and the readout selects the combination that retains signal with reduced backaction contamination. Broadband displacement experiments have used these correlations to move below the ordinary uncorrelated sensitivity curve [18, 36].

Coherent quantum noise cancellation introduces an auxiliary response with an effective sign opposite to the sensor response. Proper matching cancels backaction at the level of coherent dynamics, which is closely related to negative mass reference frames and quantum mechanics free subsystems [37].

### 6.3 Quantum nondemolition observables and backaction evasion

An ideal QND observable $\hat{A}$ satisfies

$$[\hat{A}(t_i), \hat{A}(t_j)] = 0 \tag{58}$$

at the sampling times, and the measurement interaction preserves that observable. A sufficient ideal coupling condition is

$$[\hat{A}, \hat{H}_{\mathrm{int}}] = 0. \tag{59}$$

The disturbance enters a conjugate variable whose uncertainty is absent from the later target measurement [6, 7, 12].

For a harmonic oscillator, define slowly varying rotating quadratures $\hat{X}$ and $\hat{P}$ with $[\hat{X}, \hat{P}] = i$. A balanced two tone interaction can couple the meter to one rotating quadrature. Backaction enters the conjugate quadrature, so repeated measurements of the selected quadrature evade the ordinary position SQL over the rotating wave and bandwidth regime. Counterrotating terms, damping, imperfect sideband resolution, and pump imbalance leak backaction into the measured quadrature. Experiments have implemented QND and backaction evading protocols in microwave and optical optomechanical systems [38–40].

A free mass position at a later time contains its earlier momentum, so position backaction contaminates repeated position readout. A speed meter senses a velocity like observable, and variational readout estimates a combination of output quadratures whose backaction contribution is reduced. Contractive state analyses and realizable measurement models also show that the free mass SQL depends on state preparation and measurement protocol [41, 42]. These results support careful language. Equation (20) is the SQL of the stated uncorrelated continuous position meter.

### 6.4 Phase preserving amplification

A deterministic phase preserving bosonic amplifier with power gain $G$ has the canonical map

$$\hat{a}_{\text{out}} = \sqrt{G}\,\hat{a}_{\text{in}} + \sqrt{G-1}\,\hat{b}_{\text{in}}^{\dagger}. \tag{60}$$

The auxiliary mode preserves $[\hat{a}_{\text{out}}, \hat{a}_{\text{out}}^{\dagger}] = 1$. With the auxiliary input in vacuum, the minimum input referred added noise is

$$n_{\text{add}} \geq \frac{1}{2}\left(1 - \frac{1}{G}\right), \tag{61}$$

which approaches one half quantum at large gain. Phase sensitive amplification can add zero noise to one selected quadrature as it deamplifies the conjugate quadrature. It gives up simultaneous phase preserving gain for both quadratures [10, 43].

**Table 3:** Representative strategies for surpassing a standard benchmark.

| Method | Change to the measurement model | Representative platform |
|---|---|---|
| Phase squeezing | Reduces detected phase quadrature variance | Optical phase sensors and interferometers |
| Frequency dependent squeezing | Rotates the noise ellipse across Fourier frequency | Gravitational wave detectors |
| Variational readout | Selects a correlated output quadrature that cancels backaction in the estimator | Interferometers and optomechanics |
| Backaction evasion | Couples the meter to one rotating mechanical quadrature | Microwave and optical optomechanics |
| Speed meter | Measures a velocity like observable with repeated interaction geometry | Gravitational wave detector concepts |
| Coherent noise cancellation | Matches an auxiliary response of opposite sign to the sensor response | Optomechanical force sensors |
| Spin squeezing | Reduces collective spin variance at fixed mean spin signal | Atomic clocks and magnetometers |
| Entangled probes | Increases total generator variance above separable scaling | Optical and atomic phase estimation |
| QND measurement | Selects an observable compatible with its future values | Oscillators, photons, spins, and qubits |
| Phase sensitive amplification | Amplifies one chosen field quadrature | Microwave qubit readout and radio frequency sensing |

**Practical rule of thumb**

Identify the assumption changed by each sub SQL technique. Squeezing changes the probe covariance, variational readout changes the measured quadrature, QND changes the observable, and entanglement changes generator variance.

# 7 Applications and evidence standards

**Before reading this section**

Review the difference among imprecision, added noise, and total measured noise. A claim about one level gives limited information about the others.

## 7.1 Gravitational wave interferometry

Laser interferometers infer strain from differential arm motion over kilometer baselines. Optical phase noise dominates much of the high frequency quantum noise, and radiation pressure motion gains importance at lower frequencies. Squeezed vacuum injection first improved detector sensitivity beyond the coherent shot noise level, Advanced LIGO later operated with squeezed vacuum, and frequency dependent squeezing then provided broadband quantum noise reduction [23, 44–46].

The LIGO Livingston analysis reported the quantum noise contribution below the free mass SQL from 35 to 75 Hz with a maximum improvement of 3 dB near 50 Hz after installation of the A+ frequency dependent squeezing system [47, 48]. Classical noise exceeded quantum noise in that low frequency band, so the reported quantum spectrum was inferred by calibrated model subtraction and dedicated squeeze angle measurements. The distinction preserves the importance of the result and states its experimental scope accurately.

## 7.2 Optomechanical and atomic force sensing

Murch and collaborators measured quantum radiation pressure backaction on the collective motion of an ultracold atomic gas [25]. Schreppler and collaborators observed the imprecision and backaction tradeoff in force sensing and reached a sensitivity consistent with the expected noise model at four times the absolute SQL after residual thermal disturbance and detection efficiency were included [26]. Mason and collaborators used strong optomechanical correlations to obtain off resonant total force and displacement sensitivity 1.5 dB below the SQL, including thermal and zero point motion in the total displacement spectrum [18].

Atomic force microscopy introduces cantilever thermal noise, optical readout imprecision, laser backaction, tip sample interaction noise, and bandwidth dependent calibration. Early analyses derived quantum and thermal sensitivity constraints for common AFM readouts [49, 50]. Pooser and collaborators used truncated nonlinear interferometry to measure cantilever beam displacement with up to 3 dB quantum noise reduction below the coherent optical reference and reached $1.7\,\mathrm{fm}/\sqrt{\mathrm{Hz}}$ readout sensitivity [51]. This result demonstrates quantum enhanced optical readout. A total mechanical force SQL comparison additionally requires backaction, cantilever thermal noise, and transduction calibration in the same band.

Magnetic resonance force microscopy reached single electron spin detection by coupling spin dynamics to a sensitive cantilever [52]. That milestone concerns signal detection and force sensitivity. Its relation to an SQL requires the complete quantum and thermal noise budget for the chosen force estimator.

## 7.3 Spin ensembles and clocks

For $N$ uncorrelated two level atoms polarized along one collective spin direction, transverse projection noise scales as $\sqrt{N}$ and phase uncertainty scales as $N^{-1/2}$. The Wineland metrological squeezing parameter is

$$\xi_R^2 = \frac{N(\Delta J_\perp)^2}{|\langle \mathbf{J} \rangle|^2}. \tag{62}$$

Values $\xi_R^2 < 1$ indicate phase sensitivity below the coherent spin state SQL after contrast loss is included. The parameter connects reduced collective variance with a usable signal slope and remains central to atomic clocks, magnetometers, and atom interferometers [29, 53, 54].

### 7.4 Spatial parameter estimation and quantum imaging

In quantum imaging, the estimation parameter $\theta$ is generalized to spatial coordinates $\mathbf{r}$. The resolution of a point source is not merely a consequence of diffraction, but a problem of parameter estimation where the precision is fundamentally governed by the Quantum Fisher Information $F_Q$ discussed in Section 5.

The quantum Cramér–Rao bound establishes the ultimate limit on the variance of the estimated position:

$$\mathrm{Var}(\hat{r}) \geq \frac{1}{\nu F_Q(\mathbf{r})}. \tag{63}$$

While the classical Rayleigh criterion is derived from the overlap of coherent-state point spread functions, $F_Q$ accounts for the possibility of using non-classical states to distinguish closely spaced sources. By optimizing the probe state, it is possible to redefine the resolution limit, shifting the benchmark from the diffraction limit toward the Heisenberg limit.

To achieve this in practice, the physical routes detailed in Section 6 are extended to the spatial domain. Specifically, the injection of spatially multimode squeezed vacuum or the use of entangled N00N states reduces the variance of the spatial estimator. Such techniques allow for the suppression of the classical shot-noise limit in spatial metrology, enabling super-resolution imaging where the precision of the coordinate estimate $\hat{r}$ surpasses the capabilities of any separable probe state.

**Practical rule of thumb**

When claiming super-resolution in quantum imaging, specify whether the improvement refers to the Rayleigh criterion (classical) or the quantum Cramér–Rao bound (quantum), and identify if the gain arises from the probe state (e.g., N00N states) or the measurement strategy (e.g., spatial mode sorting).

### 7.5 A hierarchy of experimental claims

**Table 5:** How the SQL enters representative platforms.

| Platform | Measured parameter | Standard balance or resource restriction | References |
|---|---|---|---|
| Laser interferometer | Differential displacement or strain | Phase imprecision against radiation pressure backaction | [9, 23, 47] |
| Cavity optomechanics | Displacement or force | Optical imprecision against photon number backaction | [14, 18] |
| Atomic force microscope | Cantilever motion and force | Optical readout, cantilever disturbance, and thermal force | [49, 51] |

| Platform | Measured parameter | Standard balance or resource restriction | References |
|---|---|---|---|
| Spin ensemble | Phase, frequency, or field | Independent projection noise against collective correlations | [53, 54] |
| Microwave amplifier | Field quadratures | Phase preserving gain against commutator preserving added noise | [10, 43] |
| Quantum phase estimation | Optical or atomic phase | Separable generator variance against correlated generator variance | [27, 32] |
| Quantum imaging | Spatial coordinate $\mathbf{r}$ | Separable spatial mode variance against entangled/squeezed modes | [55] |

**Practical rule of thumb**

Reserve the phrase total sensitivity below the SQL for a comparison that includes thermal, technical, intrinsic, imprecision, backaction, correlation, and calibration uncertainties in the same units and bandwidth.

**Table 4:** Evidence required for increasingly strong SQL statements.

| Claim | Required comparison | Common missing element |
|---|---|---|
| Sub shot noise readout | Detected readout variance against a coherent or vacuum reference at equal signal resource | Loss corrected resource accounting |
| Imprecision below SQL scale | Calibrated input referred imprecision against a stated SQL displacement scale | Backaction and thermal noise |
| Quantum noise below SQL | Calibrated sum of quantum imprecision, backaction, and correlations against the same SQL convention | Classical noise and model uncertainty |
| Total sensitivity below SQL | Measured total estimator noise, including thermal and technical contributions, against a calibrated SQL curve | Independent calibration and uncertainty propagation |
| Metrological gain below separable bound | Estimator variance or Fisher information against equal resource separable probes | State preparation cost, contrast, and bias |

# 8 Experimental design and numerical verification

**Before reading this section**

Ensure that the target parameter, estimator bandwidth, and spectral convention have already been selected. Numerical optimization is meaningful after these physical choices are fixed.

## 8.1 Noise budget and calibration sequence

A defensible SQL comparison can be built with the following sequence.

1. Define the input variable and estimator. State whether the result refers to displacement, force, strain, phase, frequency, or field.
2. Measure or model the complex transfer function from the input variable to the recorded channel. Preserve its phase.
3. Fix the Fourier normalization, spectral sidedness, symmetrization, resolution bandwidth, windowing, and averaging method.
4. Calibrate imprecision with the physical coupling suppressed or independently varied.
5. Calibrate backaction by varying probe strength and measuring the driven response or induced heating.
6. Estimate the cross spectrum with phase information. A magnitude squared coherence by itself discards the sign needed for cancellation.
7. Measure thermal and technical backgrounds and propagate their calibration uncertainty.
8. Construct the SQL from the measured effective susceptibility and the same convention used for data.

9. Repeat the comparison across probe strengths and readout quadratures to show the expected covariance geometry.

For a total measured displacement spectrum,

$$\bar{S}_{xx}^{\mathrm{meas}} = \bar{S}_{xx}^{\mathrm{intrinsic}} + \bar{S}_{xx}^{\mathrm{th}} + \bar{S}_{xx}^{\mathrm{tech}} + \bar{S}_{xx}^{\mathrm{add}}, \tag{64}$$

each component should carry a calibration covariance. Subtraction of a modeled background can reveal a quantum component, as in low frequency interferometer analyses, and the resulting statement should identify the subtraction and its uncertainty.

### 8.2 Control parameters for an SQL heat map

The analytic heat map in Fig. 4 has clear controls. The damping ratio $\gamma$ sets resonance width, the plotted frequency interval sets the mechanical response range, the range of $\log_{10}\lambda$ sets the available measurement strength, and the sample count sets rendering density. These values appear as control knobs in the preamble. A laboratory version can replace Eq. (23) with measured complex susceptibility and replace Eq. (24) with the measured covariance matrix.

Define a dimensionless excess added noise metric

$$H(\Omega,\lambda) = \log_{10}\left[\frac{\bar{S}_{xx}^{\mathrm{add}}(\Omega,\lambda)}{\bar{S}_{xx}^{\mathrm{SQL}}(\Omega)}\right]. \tag{65}$$

The meanings are direct. $H = 0$ lies on the selected SQL, $H > 0$ lies above it, and $H < 0$ lies below it. A plot with correlations must evaluate the complex cross term in Eq. (7). Using $|\bar{S}_{xF}|$ without its phase can produce a physically incorrect valley.

### 8.3 Common diagnostic failures

Several checks catch common errors before a sub SQL conclusion is drawn.

- Dimensional consistency must hold after input referral. A force spectrum multiplied by $|\chi_m|^2$ has displacement spectrum units.
- The optimum of Eq. (15) must occur where its two positive terms are equal.
- Conversion from a double sided to a one sided power spectrum multiplies by two, and the corresponding amplitude spectrum multiplies by $\sqrt{2}$.
- A squeezed variance quoted in decibels uses $10\log_{10}(V/V_0)$. An amplitude ratio uses $20\log_{10}(A/A_0)$.
- A sub SQL quantum component can coexist with total detector noise above the SQL.
- A narrowband noise minimum can differ from the setting that maximizes broadband waveform Fisher information.

**Practical rule of thumb**

Treat susceptibility phase, cross spectral phase, sidedness, and calibration covariance as primary data. These quantities decide whether a plotted sub SQL valley represents physical sensitivity or a convention mismatch.

## 9 Synthesis

The standard quantum limit gains precision once its assumptions are written beside it. An uncorrelated continuous position meter produces $\hbar|\chi_m|$ in the symmetrized double sided convention. Independent probes produce an inverse square root statistical scaling. A phase preserving amplifier adds one half quantum at large gain. Correlation aware force sensing reaches a lower dissipative constraint governed by the imaginary part of inverse susceptibility. These results share commutators, covariance matrices, and information extraction, yet they quantify distinct tasks.

Quantum enhanced sensing changes a specific part of the measurement model. Squeezing reshapes optical quadrature covariance. Variational readout retains anticorrelations in the measured record. Backaction evasion selects one rotating mechanical quadrature. Speed meters change the sensed observable. Entanglement increases generator variance. QND coupling preserves the observable needed by later measurements. Each route respects the full quantum uncertainty structure and exceeds a benchmark whose restricted assumptions have been relaxed.

The strongest experiment reports its benchmark, estimator, resource count, spectrum convention, full noise budget, and uncertainty propagation together. This practice supports valid comparisons across LIGO, optomechanics, force microscopy, spin ensembles, atomic clocks, and microwave readout.

**Practical rule of thumb**

The SQL is a calibrated design benchmark. A complete sub SQL result identifies the changed assumption and demonstrates the improvement with the full estimator noise budget.

## A Fourier and spectral conversion reference

With

$$A(\Omega) = \int_{-\infty}^{\infty} A(t)e^{i\Omega t}\,\mathrm{d}t, \qquad A(t) = \int_{-\infty}^{\infty} \frac{\mathrm{d}\Omega}{2\pi} A(\Omega)e^{-i\Omega t}, \tag{66}$$

Eq. (4) gives the double sided symmetrized spectrum. For a real stationary process,

$$\langle A^2(t)\rangle = \int_{-\infty}^{\infty} \frac{\mathrm{d}\Omega}{2\pi} \bar{S}_{AA}(\Omega) = \int_{0}^{\infty} \frac{\mathrm{d}\Omega}{2\pi} S^{(1)}_{AA}(\Omega). \tag{67}$$

Spectra per hertz use $f = \Omega/(2\pi)$ and satisfy $S^{[f]}(f)\,\mathrm{d}f = S^{[\Omega]}(\Omega)\,\mathrm{d}\Omega$. Stating the integration measure prevents a hidden factor of $2\pi$.

## B Analytic heat map definition

The heat map uses $r = \Omega/\Omega_m$, $\ell = \log_{10}\lambda$, and damping ratio $\gamma$. Its plotted function is

$$H(r,\ell) = \log_{10}\left[\frac{10^{-\ell} + 10^{\ell}|\widetilde{\chi}(r)|^2}{2|\widetilde{\chi}(r)|}\right]. \tag{68}$$

The arithmetic geometric mean inequality gives $H \geq 0$. Equality occurs at

$$\ell_{\mathrm{opt}}(r) = -\log_{10}|\widetilde{\chi}(r)|. \tag{69}$$

This analytic construction replaces the external `sql_heatmap.dat` dependency and compiles from the supplied TeX source and bibliography database.

## C Compact formula sheet

**Table 6:** Frequently used formulas and their scope.

| Formula | Scope |
|---|---|
| $\bar{S}_{xx}^{\mathrm{add}} = \bar{S}_{xx}^{\mathrm{imp}} + \lvert\chi_m\rvert^2 \bar{S}_{FF}^{\mathrm{ba}} + 2\,\mathrm{Re}(\chi_m^* \bar{S}_{xF})$ | Linear input referred displacement measurement |
| $\bar{S}_{xx}^{\mathrm{SQL}} = \hbar\lvert\chi_m\rvert$ | Ideal uncorrelated meter, symmetrized double sided spectrum |
| $\bar{S}_{FF}^{\mathrm{SQL}} = \hbar/\lvert\chi_m\rvert$ | Force referred form of the same SQL |
| $\bar{S}_{FF}^{\mathrm{add}} \geq \hbar\lvert\,\mathrm{Im}\,\chi_m^{-1}\rvert$ | Correlation aware stationary dissipative force bound |
| $F_Q = 4(\Delta G)^2$ | Pure state under unitary parameter encoding |
| $F_Q \leq \sum_j (\Delta g_j)^2$ | Separable probes with bounded local generator ranges |
| $F_Q = N^2$ | Ideal $N$ qubit GHZ probe under the stated local generator |
| $n_{\mathrm{add}} \geq (1 - G^{-1})/2$ | Deterministic phase preserving bosonic amplifier |
| $\xi_R^2 = N(\Delta J_\perp)^2/\lvert\langle \mathbf{J} \rangle\rvert^2$ | Metrological spin squeezing parameter |

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